

WASIM AKHTAR

Bengaluru, India | Open to Remote (Global) | wasimakhwork@gmail.com | Portfolio | GitHub | LinkedIn

SUMMARY

Computer Science undergraduate (AI & ML) focused on systems programming and AI developer tools. Built and open-sourced Luna, an AI terminal shell in Rust featuring persistent memory, deterministic command safety, and multi-provider LLM integration. Seeking software engineering, AI systems, and developer tools internship opportunities.

TECHNICAL SKILLS

Languages: Rust, Python, JavaScript, HTML/CSS, C/C++

AI / ML: Large Language Models (LLMs), Reinforcement Learning (PPO), Machine Learning

Systems & Tools: SQLite, Async I/O (Tokio), REST APIs, CLI Tool Design, Linux/WSL, Git/GitHub

Libraries: NumPy, Pandas, Matplotlib

PROJECTS

Luna - AI-Powered Terminal Shell | Rust, SQLite, LLM APIs | Released; Active Maintenance | GitHub

- Architected and shipped a standalone terminal shell in Rust that translates natural language into shell commands across 6 interchangeable AI providers: OpenAI, Anthropic, Gemini, Groq, Ollama, and OpenRouter.
- Designed a deterministic, rule-based risk-classification engine from CRITICAL to LOW that inspects every command before execution and blocks destructive operations by default.
- Built a SQLite-backed persistent memory layer tracking command history, directory context, and recurring error patterns across sessions to dynamically re-rank AI suggestions per user.
- Implemented an automatic error-recovery pipeline that detects failed commands, generates AI-suggested fixes, re-validates them through the safety engine, and presents them for one-click approval.
- Developed a pattern-learning module that detects repeated command sequences and converts them into reusable, replayable workflows, minimizing manual command entry.
- Shipped end-to-end across 8 build phases with a one-line install script and zero telemetry by design.

GitHub Analyzer Web App | JavaScript, HTML/CSS, REST APIs | Completed Project | Live

- Developed a web application that fetches and visualizes GitHub user statistics, including repository data and contribution graphs, using the GitHub REST API.
- Designed and implemented a custom e-ink/newspaper-style UI focused on readability and a distinctive retro user experience.
- Optimized API calls to handle rate limits efficiently while dynamically rendering user data.

RESEARCH EXPERIENCE

Student Researcher - Reinforcement Learning & Optimization | Presidency University | 2025 - Present

- Developed a PPO-based reinforcement learning framework for adaptive feature-weight optimization, replacing static MCDM weighting approaches used in prior work.
- Modeled landing-site suitability on a 5058 x 5058 lunar surface grid for autonomous exploration simulations.
- Achieved an 18-22% improvement in prediction stability and consistency over baseline methods during experimentation.
- Generated 60m-resolution suitability maps used as primary inputs for downstream simulation work.
- Currently co-authoring an academic paper detailing the methodology and experimental results under faculty guidance.

EDUCATION

Presidency University | B.Tech, Computer Science & Technology (AI & ML Specialization) | 2023 - 2027 (Expected)

Relevant Coursework: Machine Learning, Artificial Intelligence, Reinforcement Learning, Blockchain Technologies

LANGUAGES

English (Fluent), Hindi (Fluent)